

Algorithmic Decision Making Framework

INTRODUCTION

We introduce readers to the algorithmic decision making framework. The process includes macro- and micro-level decisions specified prior to trading to ensure consistency between the “the trading goal” and the “investment objective.” The macro-level decision refers to the best execution strategy (BES) most consistent with the investment objective. The micro-level decision refers to how the algorithm will behave in real-time and how it will adapt to changing market conditions. Subsequently, it is the goal of the limit order model and smart order router to ensure that order placement and actual executions adhere to the investor’s specifications. Only investors who possess full knowledge and proper specification of these criteria will be positioned to achieve best execution.

Before we discuss our algorithmic decision making framework, it is important to restate a few important concepts. Algorithmic trading is the computerized execution of financial instruments following pre-specified rules and guideline. Algorithmic trading provides many benefits. They do exactly what they are instructed to do—and do it well. However, one of the more unfortunate aspects of algorithmic trading is that they do exactly what they are instructed to do. If they are not provided with instructions that are in the best interest of the fund over all possible sets of market events, the results will likely be unfavorable execution and subpar performance.

Algorithmic decision frameworks have been previously studied from the perspective of the macro and micro viewpoint. For example, macro decisions have been studied by Barra (1997); Bertsimas and Lo (1998); Almgren and Chriss (1999, 2000); Cox (2000); and Kissell, Glantz, and Malamut (2004). Micro decisions have been studied more recently such as in Journal of Trading’s “Algorithmic Decision Making Framework,”